PCH 202 — Inorganic Pharmaceutical Chemistry

Practice Workbook & Answer Key | ABUAD 200 Level Pharmacy | Second Semester

An original revision aid by the EverythingABUAD student team | everythingabuad.com

How to use this workbook. The questions below are grouped by topic and were written from scratch by our team to help you revise the core concepts of Inorganic Pharmaceutical Chemistry — volumetric analysis, bonding, electronic effects, periodicity, coordination chemistry, nomenclature and pharmaceutical assays. They are original practice items for self-testing — not past exam papers and not a prediction of any future test. Try each question on your own first, then check yourself against the worked answer key at the end. Always cross-check with your lecturer's current course outline.

Volumetric (Titrimetric) Analysis

1. Define titrimetric analysis and name the two solutions that are always involved in a titration.

2. List four broad classes of titration used in pharmaceutical analysis and give the reaction each is based on.

3. Distinguish clearly between the equivalence point and the end point of a titration.

4. Define a primary standard and state four properties a substance must have to qualify as one.

5. Differentiate between a primary standard and a secondary standard, giving one example of each.

6. Explain why sodium thiosulphate solution must be standardised before it is used to assay other substances.

7. What is the role of potassium iodate as a primary standard in the standardisation of sodium thiosulphate?

8. Why is starch mucilage added only near the end point of an iodine/thiosulphate titration and not at the start?

9. A student standardises sodium thiosulphate against iodine liberated from 0.4280 g of KIO_3 (M = 214 g/mol) made up and titrated in portions, and 20.0 mL of thiosulphate is required for the iodine from a 1/10 aliquot. Outline the calculation of the thiosulphate concentration. (1 mol KIO_3 liberates I_2 equivalent to 6 mol thiosulphate.)

10. Define redox titration and identify the species oxidised and reduced when potassium permanganate is titrated against oxalic acid.

11. Balance the reaction $MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$ in acidic medium.

12. 25.00 mL of an oxalic acid solution reacts with 18.40 mL of 0.05 M $KMnO_4$. Using the 2:5 ratio, calculate the mass of oxalic acid in the 100 mL from which the 25 mL was taken. (M of $C_2H_2O_4$ = 90.03 g/mol.)

13. Why must potassium permanganate itself be standardised rather than used as a primary standard?

14. Why is no external indicator needed in a permanganate titration?

15. What is a blank titration and why is it performed?

16. Given that 0.027801 g of a sample is equivalent to 1 mL of 0.1 M titrant (factor = 0.9677) and the average titre is 24.20 mL, outline how you would calculate the percentage purity of the sample.

17. State five precautions that should be observed to obtain accurate and reproducible titration results.

18. Why is distilled water 'recently boiled and cooled' before being used in certain acid-base titrations?

Solution Concentration and Calculations

19. Differentiate between molarity and normality.

20. The glucose concentration in normal blood is about 90 mg per 100 mL. Express this as a molar concentration. (M of glucose, $C_6H_{12}O_6$ = 180 g/mol.)

21. Describe how you would prepare 500 mL of 0.100 M sodium hydroxide solution from solid NaOH pellets. (M of NaOH = 40 g/mol.)

22. How many grams of potassium iodate are needed to prepare 250 mL of 0.0200 M KIO_3? (M = 214 g/mol.)

23. Convert 0.05 M H_2SO_4 to normality, and explain the difference.

Chemical Bonding

24. Compare and contrast ionic, covalent and coordinate (dative) bonds in terms of how the bonding electrons are supplied.

25. Contrast the typical physical properties expected for an ionic compound versus a covalent compound.

26. Distinguish between a sigma (σ) bond and a pi (π) bond.

27. State the three basic mechanisms of bond fission and describe the species each produces.

28. Explain how VSEPR theory predicts the shape of a molecule, and use it to predict the shape and bond angle of methane, ammonia and water.

29. Explain what hybridisation means and state the geometry and bond angle associated with sp, sp^2 and sp^3 hybrid orbitals.

30. For the molecule $HC{\equiv}C{-}CH_2{-}NH{-}CH{=}CH_2$, give the hybridisation, geometry and approximate bond angle of a triple-bonded carbon, the CH_2 carbon, and a double-bonded carbon.

31. Why is a single Lewis (dot-and-cross) structure inadequate to describe a molecule such as aniline or the nitrate ion?

Electronic Effects and Reactivity

32. Differentiate between the inductive effect and the mesomeric (resonance) effect.

33. Define electronegativity and describe how it varies across a period and down a group.

34. Explain what is meant by the 'differential electronegativity' of a carbon atom and why it matters in reactions.

35. Arrange the carbocations $^{+}CH_3$, $^{+}CH(CH_3)_2$ and $^{+}C(CH_3)_3$ from least to most stable and explain the order.

36. Using aniline as an example, describe how a substituent can induce a separation of charge (polarisation) in a molecule.

37. Distinguish between homolytic and heterolytic cleavage and state the reaction conditions that favour each.

Periodicity and Atomic Structure

38. State the modern periodic law.

39. Name the four quantum numbers, state what each describes and give the allowed values of each.

40. State Hund's rule of maximum multiplicity.

41. Explain, with a relevant example, why atomic radius increases down a group but decreases across a period.

42. Classify the elements of the periodic table into their four broad types.

43. A sample of urea, $CO(NH_2)_2$, should theoretically contain 46.65% nitrogen but analysis finds 40.0%. Calculate the theoretical value and state what the result indicates. (M: C=12, O=16, N=14, H=1.)

Coordination Chemistry

44. Define the terms central atom, ligand and coordination sphere in a coordination compound.

45. Classify ligands by the number of donor atoms they use, giving an example of each, and state what an example of a neutral, an anionic and a unidentate ligand is.

46. Define coordination number and give the coordination number of cobalt in $[Co(NH_3)_6]^{3+}$.

47. Name the coordination compounds (i) $[CoCl(NH_3)_5]Cl_2$ and (ii) $K_4[Fe(CN)_6]$, and identify the ligands and coordination sphere in each.

48. Define chelation and give two examples of chelating agents used in pharmacy.

49. Why is the ability of EDTA to form chelates important in complexometric analysis, and state one pharmaceutical use of EDTA.

50. Distinguish between the coordination sphere and the ionisation sphere of a complex, using $[CoCl(NH_3)_5]Cl_2$ as an example.

IUPAC Nomenclature

51. State the IUPAC rule used to name a compound that contains both a double and a triple bond, and name $HC{\equiv}C\text{-}CH_2\text{-}CH_2\text{-}CH{=}CH_2$ on that basis.

52. Give the IUPAC name of $CH_3\text{-}C{\equiv}C\text{-}H$.

53. Give the IUPAC name of the branched alkane $CH_3CH(CH_3)(CH_2)_3CH(CH_2CH_3)$... i.e. a heptane chain bearing a methyl and an ethyl substituent.

54. Provide IUPAC names for a benzene ring bearing -COOH, -Cl and $-CH_3$ (a chloro-methyl-benzoic acid).

55. Name the alkanes of chain length 1, 4, 6, 11 and 20 carbons.

56. Circle-and-name any five different functional groups present in a molecule bearing an amide, a nitrile, a thiol, an aryl bromide and an ether.

Pharmaceutical Inorganic Chemistry and Antacids

57. State three general types of chemical reaction undergone by Group IIA (alkaline-earth) elements, writing a balanced equation for one.

58. Explain the rationale behind combining aluminium hydroxide with magnesium trisilicate (or magnesium hydroxide) in antacid preparations.

59. Why is magnesium sulphate not used as an antacid even though it is a magnesium compound?

60. Explain the rationale for formulating dried aluminium hydroxide and magnesium hydroxide as a Gestid-type suspension.

61. What is the major rationale for adding a flavour/other agent, and generally for combining ingredients, in an antacid preparation to improve patient acceptability?

62. Give a balanced equation for the neutralisation of stomach acid by aluminium hydroxide and by magnesium hydroxide.

Assays and Quality Control

63. Define an assay and state the difference between a chemical assay and a biological assay (bioassay).

64. State four characteristics of a good assay method.

65. Briefly explain the principle of microbiological assays and name two methods used.

66. Enumerate at least five advantages of high-performance liquid chromatography (HPLC) as a technique for chemical assay.

67. Enumerate and explain the causes of poor-quality products in quality assurance.

68. Write short notes on: (i) competitive immunoassay, (ii) quality control, and (iii) gross errors.

69. Briefly discuss retention time and quantal bioassay.

Answer Key

Worked answers, grouped to match the questions above.

Volumetric (Titrimetric) Analysis

1. Titrimetric (volumetric) analysis measures the volume of a reagent of known concentration needed to react completely with a measured amount of the analyte. The two solutions are the **titrant** (standard solution in the burette) and the **analyte** (the solution of unknown concentration in the flask).

2. **Acid-base** (neutralisation, H^+ + OH^-); **redox** (electron transfer, e.g. permanganate vs oxalate); **precipitation** (formation of an insoluble salt, e.g. argentometry with $AgNO_3$); **complexometric** (formation of a soluble complex, e.g. EDTA with metal ions).

3. The **equivalence point** is the theoretical point at which the titrant added is stoichiometrically equal to the analyte. The **end point** is the experimentally observed point (usually a colour change) taken to signal completion. A well-chosen indicator makes the two nearly coincide; the small difference is the **titration (indicator) error**.

4. A **primary standard** is a highly pure, stable substance that can be weighed directly to prepare a solution of accurately known concentration. It must: (1) be of high, known purity; (2) be stable in air and not hygroscopic or efflorescent; (3) have a high molar mass (to minimise weighing error); and (4) react rapidly and stoichiometrically with the analyte. Examples: sodium carbonate, potassium hydrogen phthalate, potassium iodate.

5. A **primary standard** (e.g. potassium iodate, KIO_3) is pure and stable enough to make an exact-concentration solution by direct weighing. A **secondary standard** (e.g. sodium thiosulphate, $Na_2S_2O_3$) is not stable or pure enough for this and must be *standardised* against a primary standard before use.

6. Sodium thiosulphate is a secondary standard: it is efflorescent, its solutions slowly decompose (by dissolved CO_2, air oxidation and bacteria), so its exact concentration drifts over time. It must therefore be standardised against a freshly prepared primary standard such as potassium iodate close to the time of use.

7. Potassium iodate (KIO_3) is weighed accurately and reacted with excess acidified potassium iodide to liberate a stoichiometric amount of iodine (IO_3^- + $5I^-$ + $6H^+$ -> $3I_2$ + $3H_2O$). The liberated iodine is then titrated with the thiosulphate, so the exactly known mass of KIO_3 fixes the true concentration of the thiosulphate.

8. Starch forms an intense blue complex with iodine. If added while iodine is still concentrated, it binds a large amount of iodine so tightly that release is slow and the end point becomes sluggish and inaccurate. Added near the end (pale straw colour), only a little iodine remains, so the blue-to-colourless change is sharp and reversible.

9. Moles KIO_3 in the aliquot = (0.4280 / 214) x (1/10) = 2.00 x 10^{-4} mol. Each mole of iodate is equivalent to 6 moles of thiosulphate, so moles $S_2O_3^{2-}$ = 6 x 2.00 x 10^{-4} = 1.20 x 10^{-3} mol. Concentration = moles / volume = 1.20 x 10^{-3} / 0.0200 L = **0.060 M**. General method: exact conc = (6 x moles KIO_3) / titre volume in litres.

10. A **redox titration** is based on transfer of electrons between the titrant and analyte. With $KMnO_4$ and oxalic acid: manganese in MnO_4^- is **reduced** (Mn^{+7} -> Mn^{+2}, gaining 5 e^-) and carbon in $C_2O_4^{2-}$ is **oxidised** (to CO_2, losing electrons). $KMnO_4$ is the oxidising agent, oxalic acid the reducing agent.

11. $2\ MnO_4^- + 5\ C_2O_4^{2-} + 16\ H^+ \rightarrow 2\ Mn^{2+} + 10\ CO_2 + 8\ H_2O$. Check: Mn 2=2; C 10=10; charge left (-2-10+16 = +4) = right (+4); O and H balance. The 2:5 ratio reflects 5 e^- gained per MnO_4^- and 2 e^- lost per oxalate.

12. Moles $KMnO_4$ = 0.05 x 0.01840 = 9.20 x 10^{-4} mol. From 2 MnO_4 : 5 oxalate, moles oxalate = (5/2) x 9.20 x 10^{-4} = 2.30 x 10^{-3} mol in the 25 mL. In 100 mL: 2.30 x 10^{-3} x (100/25) = 9.20 x 10^{-3} mol. Mass = 9.20 x 10^{-3} x 90.03 = **0.828 g**.

13. Commercial $KMnO_4$ is not pure enough: it is contaminated with MnO_2, and its solutions are slowly reduced by traces of organic matter and by light, lowering the concentration over time. It is therefore a secondary standard and is standardised (usually against sodium oxalate or oxalic acid) shortly before use.

14. Permanganate is intensely purple, while Mn^{2+} is almost colourless. As long as analyte remains, added permanganate is decolourised; the end point is the first faint permanent pink from one drop of excess $KMnO_4$. It is therefore a **self-indicating** titrant.

15. A **blank titration** is carried out on all the reagents (solvent, acid, indicator) *without* the analyte, under identical conditions. It measures the volume of titrant consumed by impurities, the indicator, or dissolved gases. Subtracting the blank from the sample titre corrects for these and improves accuracy.

16. Corrected titre = observed titre x factor = 24.20 x 0.9677 = 23.42 mL of exactly 0.1 M titrant. Mass reacted = 23.42 x 0.027801 = 0.6512 g. Percentage purity = (mass found / mass weighed) x 100. So % purity = (0.6512 / weighed mass) x 100; if 0.65 g was weighed, purity ~= 100%. The factor scales the nominal molarity to the true molarity.

17. (1) Rinse the burette and pipette with the solutions they will hold to avoid dilution; (2) remove air bubbles from the burette tip and read the meniscus at eye level to avoid parallax; (3) add titrant slowly, with swirling, near the end point; (4) run determinations in duplicate (or to concordant titres within 0.1 mL); (5) use freshly boiled and cooled distilled water where CO_2 would interfere, and add the indicator in the correct, small amount.

18. Boiling expels dissolved carbon dioxide, which would otherwise form carbonic acid and neutralise part of an alkaline titrant (or shift the end point), introducing error. Cooling before use prevents evaporation losses and volume error from a hot solution.

Solution Concentration and Calculations

19. **Molarity (M)** = moles of solute per litre of solution. **Normality (N)** = number of equivalents of solute per litre, where equivalents = moles x n (n = H^+/OH^- exchanged in acid-base, or electrons transferred in redox). Thus N = M x n; for a monoprotic acid N = M, but for H_2SO_4 (n = 2), N = 2M.

20. 90 mg/100 mL = 0.90 g/L = 900 mg/L. Moles per litre = 0.90 / 180 = **5.0 x 10^{-3} mol/L (5.0 mM)**.

21. Mass needed = M x V x molar mass = 0.100 x 0.500 x 40 = **2.00 g**. Weigh 2.00 g of NaOH pellets, dissolve in about 300 mL of recently boiled, cooled distilled water in a beaker, transfer quantitatively into a 500 mL volumetric flask, rinse the beaker into the flask, then make up to the 500 mL mark and mix. (Since NaOH is a secondary standard, standardise against a primary standard such as KHP before accurate use.)

22. Mass = M x V x molar mass = 0.0200 x 0.250 x 214 = **1.07 g**. Weigh accurately, dissolve, and make up to 250 mL in a volumetric flask; because KIO_3 is a primary standard, this gives an exactly known concentration.

23. For H_2SO_4, n = 2 (two ionisable H^+). Normality = M x n = 0.05 x 2 = **0.10 N**. Molarity counts molecules per litre; normality counts reactive equivalents, which is why a diprotic acid has twice the normality of its molarity.

Chemical Bonding

24. In an **ionic bond**, electrons are transferred from one atom to another, giving oppositely charged ions held by electrostatic attraction (e.g. NaCl). In a **covalent bond**, two atoms each contribute one electron to a shared pair (e.g. Cl_2). In a **coordinate (dative) bond**, both shared electrons come from a single atom, the donor (e.g. the N->B bond in $H_3N \rightarrow BF_3$, or metal-ligand bonds). Once formed, a coordinate bond is indistinguishable from an ordinary covalent bond.

25. **Ionic** compounds are usually high-melting crystalline solids, hard and brittle, conduct electricity when molten or dissolved, and are often soluble in water. **Covalent (molecular)** compounds are usually low-melting solids, liquids or gases, do not conduct electricity, and are more soluble in non-polar solvents. The differences arise because ionic lattices need large energy to break many electrostatic bonds, whereas molecular solids are held by weaker intermolecular forces.

26. A **sigma bond** forms by end-on (head-on) overlap of orbitals along the internuclear axis, giving electron density concentrated between the nuclei; it is strong and allows free rotation. A **pi bond** forms by side-on overlap of parallel p orbitals, with density above and below the axis; it is weaker and prevents rotation. A single bond is one σ; a double bond is one σ + one π; a triple bond is one σ + two π.

27. **Homolytic fission**: the bond breaks evenly, each atom keeping one electron, producing two **free radicals** (e.g. $Cl^{\cdot}$). **Heterolytic fission**: the bond breaks unevenly, one atom keeping both electrons, producing a **cation and an anion** (e.g. a carbocation + a leaving anion). (A third descriptive type, **dative fission**, is the reverse of coordinate bond formation, returning both electrons to the donor.)

28. VSEPR (valence-shell electron-pair repulsion) states that electron pairs around a central atom arrange themselves as far apart as possible to minimise repulsion; lone pairs repel more strongly than bonding pairs. **CH_4**: 4 bonding pairs -> tetrahedral, 109.5°. **NH_3**: 3 bonding + 1 lone pair -> trigonal pyramidal, ~107°. **H_2O**: 2 bonding + 2 lone pairs -> bent, ~104.5°. Bond angle falls as lone pairs increase.

29. **Hybridisation** is the mixing of atomic orbitals of similar energy to form equivalent hybrid orbitals directed for bonding. **sp**: 2 hybrids, linear, 180° (e.g. alkyne carbon). **sp^2**: 3 hybrids, trigonal planar, 120° (e.g. alkene carbon). **sp^3**: 4 hybrids, tetrahedral, 109.5° (e.g. alkane carbon).

30. Triple-bonded (alkyne) carbon: **sp, linear, 180°**. The saturated CH_2 carbon: **sp^3, tetrahedral, 109.5°**. Double-bonded (alkene) carbon: **sp^2, trigonal planar, 120°**. Hybridisation is set by the number of σ bonds + lone pairs on the atom.

31. Because the real molecule is a **resonance hybrid**: the bonding electrons (particularly π electrons and lone pairs) are **delocalised** over several atoms, so no single Lewis structure shows the true, averaged electron distribution. In nitrate (NO_3^-) all three N-O bonds are identical (bond order 1■), and in aniline the nitrogen lone pair is delocalised into the ring; a single structure wrongly implies fixed single and double bonds.

Electronic Effects and Reactivity

32. The **inductive effect** is the permanent polarisation of a σ bond due to electronegativity differences; it is transmitted through bonds, weakens rapidly with distance, and can be electron-withdrawing (-I) or electron-donating (+I). The **mesomeric (resonance) effect** is the shift of π or lone-pair electrons through a conjugated system; it acts over the whole conjugated framework and can be +M (donating) or -M (withdrawing).

33. **Electronegativity** is the tendency of an atom to attract the shared pair of electrons in a bond toward itself. It **increases across a period** (nuclear charge rises, radius falls) and **decreases down a group** (added shells increase the radius and shielding). Fluorine is the most electronegative element.

34. It refers to the fact that a carbon atom's effective electronegativity is not fixed but depends on its hybridisation and neighbours: an sp carbon (more s-character) is more electronegative than sp^2, which is more than sp^3. This variation controls bond polarity, acidity (terminal alkyne H is weakly acidic) and the sites at which nucleophiles or electrophiles attack.

35. Least to most stable: **$^+CH_3$ (methyl) < $^+CH(CH_3)_2$ (secondary) < $^+C(CH_3)_3$ (tertiary)**. Alkyl groups are electron-donating (+I) and also stabilise the positive charge by hyperconjugation; the more alkyl groups attached to the charged carbon, the more the charge is dispersed, so the tertiary cation is the most stable.

36. In aniline the nitrogen lone pair is delocalised into the benzene ring by the +M (mesomeric) effect, so negative charge builds up at the ortho and para ring positions while the nitrogen bears partial positive charge. This charge separation makes those ring positions electron-rich and activates them toward electrophilic attack; it also explains aniline's lower basicity than an aliphatic amine, since the lone pair is partly tied up in the ring.

37. **Homolytic cleavage** splits a bond so each fragment keeps one electron, giving radicals; it is favoured by non-polar conditions, heat or UV light (e.g. halogenation of alkanes). **Heterolytic cleavage** splits the bond so one fragment keeps both electrons, giving ions; it is favoured by polar solvents and polar bonds (e.g. ionisation of an alkyl halide to a carbocation).

Periodicity and Atomic Structure

38. The modern periodic law states that the **physical and chemical properties of the elements are periodic functions of their atomic numbers** (not atomic masses, as in Mendeleev's original law).

39. **Principal (n)**: main energy level/size; n = 1,2,3,... **Azimuthal/angular (l)**: subshell/shape; l = 0 to n-1 (s,p,d,f). **Magnetic (m_l)**: orbital orientation; m_l = -l ... 0 ... +l. **Spin (m_s)**: electron spin; m_s = +½ or -½.

40. **Hund's rule**: when filling orbitals of equal energy (degenerate orbitals), electrons occupy them singly and with parallel spins first, before any orbital is doubly occupied. This minimises electron-electron repulsion and gives the most stable (lowest-energy) arrangement.

41. **Down a group** extra electron shells are added, so the outer electrons are farther from the nucleus and more shielded: radius increases (e.g. Li < Na < K). **Across a period** electrons enter the same shell while nuclear charge rises, pulling the shell inward: radius decreases (e.g. Na > Mg > Al). Example proving the group trend: the atomic radius rises steadily from Li (~152 pm) to Cs.

42. **Metals** (lose electrons, form cations; most elements), **non-metals** (gain/share electrons, upper right), **metalloids/semi-metals** (intermediate properties, along the staircase, e.g. Si, As), and by block: **s-, p-, d- (transition) and f- (inner transition) block** elements. Elements are also grouped as representative (main-group), transition and noble gases.

43. Molar mass of urea = 12 + 16 + 2(14+2) = 60 g/mol. Nitrogen mass = 2 x 14 = 28. Theoretical %N = (28/60) x 100 = **46.67%**. Since the found value (40.0%) is lower than the theoretical 46.67%, the sample is **impure** (contains non-nitrogenous impurity or has partly decomposed).

Coordination Chemistry

44. The **central atom** is the metal ion that accepts electron pairs. A **ligand** is the ion or molecule that donates a lone pair to the metal (a Lewis base). The **coordination sphere** is the central metal together with its directly bonded ligands, written in square brackets; ions outside the brackets form the **ionisation sphere**.

45. By donor atoms: **unidentate/monodentate** (one donor, e.g. NH_3, Cl^-), **bidentate** (two, e.g. ethylenediamine), **polydentate/chelating** (several, e.g. EDTA, hexadentate). NH_3 is a **neutral, unidentate** ligand; Cl^- is an **anionic, unidentate** ligand.

46. The **coordination number** is the number of donor atoms (ligand bonds) directly attached to the central metal ion. In $[Co(NH_3)_6]^{3+}$ six ammonia ligands are bonded, so the coordination number is **6**.

47. (i) **Pentaamminechloridocobalt(III) chloride**; ligands = 5 NH_3 + 1 Cl^-, coordination sphere = $[CoCl(NH_3)_5]^{2+}$, Co is +3. (ii) **Potassium hexacyanidoferrate(II)**; ligands = 6 CN^-, coordination sphere = $[Fe(CN)_6]^{4-}$, Fe is +2. Ligands are named alphabetically before the metal; anionic complexes take the suffix -ate.

48. **Chelation** is the formation of a ring structure when a polydentate ligand binds a metal ion through two or more donor atoms. Pharmaceutical chelating agents include **EDTA** (and its salts, used to treat metal poisoning and as a stabiliser) and **dimercaprol / penicillamine / desferrioxamine** (for lead, copper and iron overload respectively).

49. EDTA is hexadentate and forms very stable 1:1 chelates with almost all metal ions, so a single sharp end point (with a metal indicator) allows accurate determination of metal-ion content in complexometric titrations. Pharmaceutically, EDTA (as calcium disodium or disodium salt) is used to **treat heavy-metal (e.g. lead) poisoning** and as an antioxidant/stabilising chelator in

formulations.

50. The **coordination sphere** $[CoCl(NH_3)_5]^{2+}$ holds ligands bonded directly to the metal and does not dissociate in solution. The **ionisation sphere** is the counter-ions outside the brackets (the two Cl^-), which dissociate freely and can be precipitated (e.g. by $AgNO_3$). Only the ionisation-sphere chloride is precipitated, which is how the two are distinguished experimentally.

IUPAC Nomenclature

51. When both a double and triple bond are present, the chain is numbered to give the **lowest set of locants** to the multiple bonds as a set; if there is a choice, the **double bond gets the lower number**. The compound is named as an '-en-yne'. $HC\equiv C-CH_2-CH_2-CH=CH_2$ (6 carbons) = **hex-1-en-5-yne** (numbering from the double-bond end).

52. This is a three-carbon chain with a terminal triple bond: **propyne (prop-1-yne)**.

53. Identify the longest chain, number it for lowest locants, and cite substituents alphabetically. For a heptane parent carrying an ethyl group and a methyl group, the name has the form **x-ethyl-y-methylheptane**, e.g. **4-ethyl-2-methylheptane** (exact locants depend on the drawn positions; ethyl is cited before methyl alphabetically).

54. The -COOH group is the principal characteristic group, so the ring is named as a **benzoic acid** with COOH at C1; the chloro and methyl substituents are numbered for lowest locants and cited alphabetically, e.g. **2-chloro-4-methylbenzoic acid**.

55. 1 = **methane**; 4 = **butane**; 6 = **hexane**; 11 = **undecane**; 20 = **eicosane (icosane)**.

56. Five distinct functional groups: **amide** (-C(=O)-NH-), **nitrile** (-C≡N), **thiol** (-SH), **halide/aryl bromide** (Ar-Br) and **ether** (-O-). Each is identified by its characteristic atom grouping regardless of the rest of the skeleton.

Pharmaceutical Inorganic Chemistry and Antacids

57. (1) With oxygen -> basic oxides: **$2 Mg + O_2 \rightarrow 2 MgO$**. (2) With water -> hydroxide + hydrogen: $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$. (3) With acids -> salt + hydrogen: $Mg + 2HCl \rightarrow MgCl_2 + H_2$. They generally act as reducing agents, losing two electrons to form M^{2+}.

58. Aluminium salts are **constipating** and act slowly, whereas magnesium salts are **laxative** and act quickly. Combining them **cancels the opposing effects on the bowel** and gives a smoother, more sustained acid-neutralising action, improving patient tolerability.

59. Because magnesium sulphate is a **saline purgative (laxative)**: the sulphate ion is poorly absorbed and draws water into the gut, causing catharsis rather than acid neutralisation. Antacid magnesium compounds are the hydroxide, carbonate and trisilicate, which react with gastric acid without a strong purgative effect.

60. A suspension keeps the insoluble hydroxides finely and uniformly dispersed, giving a large surface area for rapid acid neutralisation and an easily swallowed, accurately dosed liquid. Combining the constipating aluminium hydroxide with the laxative magnesium hydroxide balances the bowel effects, while the suspension form coats and soothes the gastric mucosa.

61. To **improve the taste and palatability** of an otherwise chalky, unpalatable medicine, thereby increasing patient compliance, and to balance side-effects (e.g. bowel effects) so the product is better tolerated over repeated dosing.

62. Aluminium hydroxide: **$Al(OH)_3$ + 3HCl -> $AlCl_3$ + $3H_2O$**. Magnesium hydroxide: **$Mg(OH)_2$ + 2HCl -> $MgCl_2$ + $2H_2O$**. Both consume gastric HCl, raising stomach pH.

Assays and Quality Control

63. An **assay** is an investigative procedure for qualitatively detecting or quantitatively measuring the amount or activity of a target substance (the analyte). A **chemical assay** measures the amount (concentration/purity) of a substance by chemical/physicochemical means (e.g. titration, HPLC). A **biological assay** measures the biological activity or potency of a substance by its effect on a living system (organism, tissue or cells), used when activity cannot be fully captured chemically.

64. A good assay should be **specific/selective** (measures only the analyte), **accurate and precise** (close to the true value and reproducible), **sensitive** (detects small amounts), and **robust, rapid and economical** (reliable across small changes, quick and affordable). Linearity over the working range is also expected.

65. Microbiological assays measure a substance (e.g. an antibiotic or vitamin) by its effect on the growth of a sensitive micro-organism; the response is compared with a standard to estimate potency. Two methods: the **cylinder-plate (agar diffusion) method**, where zone-of-inhibition diameter is measured, and the **turbidimetric (tube) method**, where growth inhibition is measured as turbidity.

66. (1) High resolution/separating power for complex mixtures; (2) high sensitivity (detects trace amounts); (3) good accuracy and reproducibility; (4) rapid analysis; (5) applicable to non-volatile and thermally labile compounds (unlike GC); (6) can be automated and coupled to various detectors for both qualitative and quantitative work.

67. Causes include: **substandard raw materials** (impure or wrong ingredients); **faulty manufacturing processes/equipment** (poor mixing, contamination); **inadequate process controls and validation**; **poor storage/packaging** leading to degradation; **human error and poor training**; and **weak or absent quality-control testing** so defects go undetected. Each undermines the identity, purity, potency or stability of the finished product.

68. (i) **Competitive immunoassay**: labelled and unlabelled antigen compete for a limited amount of antibody; the measured signal is inversely related to the amount of analyte. (ii) **Quality control**: the part of quality assurance concerned with sampling, specification and testing to ensure a product meets required standards before release. (iii) **Gross errors**: large, obvious mistakes (e.g. wrong weighing, spillage, misread burette) that make a result unusable and are not treated statistically but are simply rejected/repeated.

69. **Retention time** is the time between injection and the appearance of a peak maximum in chromatography; under fixed conditions it is characteristic of a compound and used for identification. **Quantal (all-or-none) bioassay** records an all-or-nothing response (e.g. death, protection, or a defined effect) in a group of subjects and relates the proportion responding to dose, from which potency (e.g. ED_{50}, LD_{50}) is estimated.